NUMERICAL EVIDENCE FOR CONJECTURES ON NEGATIVE DISCRETE SECOND MOMENTS OF DIRICHLET L-FUNCTIONS

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Abstract. Pearce-Crump recently conjectured that, for a primitive Dirichlet character χ modulo q, $\sum_{0<\gamma\le T}|L'(\rho,\chi)|^{-2}\sim\frac{A_q}{2\pi}T$ with $A_q=\frac{6}{\pi^2}\prod_{p\mid q}\frac{p}{p+1}$, and $\sum_{0<\gamma\le T}|L(2\rho,\chi^2)/L'(\rho,\chi)|^2\sim\frac{\varphi(q)}{q}\frac{T}{2\pi}$. These conjectures come from a long-mollifier heuristic and generalise conjectures of Gonek and Ng for the Riemann zeta-function. We test them numerically for the real primitive characters of conductor 4, 5, 8 and 12, using between 2000 and 3000 zeros of each L-function, computed and verified for completeness, together with the first 27000 zeros of $\zeta(s)$, comparing each L-function with $\zeta(s)$ at the same local density of zeros. In all eight tests the predicted arithmetic factor lies inside the 95% bootstrap interval and the value without arithmetic factor is excluded. For conductor 12, where the two conjectures predict $1/2$ and $1/3$, we measure $0.501\,[0.445,0.555]$ and $0.340\,[0.301,0.381]$. All predictions and analysis choices were fixed before the corresponding computations. We also observe that the recipe of the ratios conjecture reproduces both constants with an error term $O(T^{1/2+\varepsilon})$, so that no lower-order terms of size $T/\log T$ are expected; the deviations seen at low height are consistent with the heavy tails of the summands.

1. Introduction

Let $\rho=\frac{1}{2}+i\gamma$ denote the non-trivial zeros of $\zeta(s)$. Assuming the Riemann Hypothesis and that all zeros are simple, Gonek [4] conjectured

$$\sum_{0<\gamma\le T}\frac{1}{|\zeta'(\rho)|^2}\sim\frac{3}{\pi^3}\,T, \tag{1}$$

and Milinovich and Ng [7] proved a lower bound of half this size. Ng [8] conjectured the companion asymptotic $\sum_{0<\gamma\le T}|\zeta(2\rho)/\zeta'(\rho)|^2\sim T/2\pi$, with a matching lower bound of half the size due to Sinha [10]. Negative discrete moments of this kind enter explicit formulae for the summatory functions of the Möbius and Liouville functions, and their sizes were modelled with random matrix theory by Hughes, Keating and O'Connell [6]. Numerically, Hiary and Odlyzko [5] found agreement with (1) using large sets of zeros near the 10^{16}th, 10^{20}th and 10^{23}rd zeros.

For a primitive Dirichlet character χ modulo $q>1$, Pearce-Crump [9] proved lower bounds for

$$J_\chi(T)=\sum_{0<\gamma\le T}\frac{1}{|L'(\rho,\chi)|^2},\qquad K_\chi(T)=\sum_{0<\gamma\le T}\left|\frac{L(2\rho,\chi^2)}{L'(\rho,\chi)}\right|^2,$$

uniformly in the conductor, and formulated the following conjectures [9, Conjectures 7 and 8]:

$$J_\chi(T)\sim\frac{A_q}{2\pi}\,T,\qquad A_q=\frac{6}{\pi^2}\prod_{p\mid q}\frac{p}{p+1}, \tag{2}$$

$$K_\chi(T)\sim\frac{\varphi(q)}{q}\,\frac{T}{2\pi}. \tag{3}$$

Both reduce to the conjectures of Gonek and Ng when $q=1$. They were obtained from Farmer's long-mollifier heuristic [3], which, as Pearce-Crump points out, is not a proof and is applied there to

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discrete sums rather than to the continuous mean values for which it was designed. Their derivation from the ratios conjecture [1, 2] is left open in [9].

The content of (2) and (3) beyond the case $q = 1$ lies in the arithmetic factors $\prod_{p|q} p/(p+1)$ and $\varphi(q)/q$. The aim of this note is to test these factors numerically. We are not aware of previous numerical tests of (2) or (3); our literature search is described in Section 8. We also record two heuristic checks: the random matrix model of Hughes, Keating and O'Connell gives the same constants (Section 5), and so does the recipe of the ratios conjecture, which moreover predicts an error term $O(T^{1/2+\varepsilon})$ (Section 6).

2. Method

2.1. **Normalisation.** Since the zeros of $L(s,\chi)$ at height t have density $\frac{1}{2\pi}\log\frac{qt}{2\pi}$, (2) and (3) say, to first order, that the zero-wise averages of

$$v_n = \frac{\log(q\gamma_n/2\pi)}{|L'(\rho_n,\chi)|^2}, \qquad w_n = \frac{\log(q\gamma_n/2\pi)\,|L(2\rho_n,\chi^2)|^2}{|L'(\rho_n,\chi)|^2}$$

tend to A_q and $\varphi(q)/q$ respectively (to $6/\pi^2$ and 1 for ζ).

2.2. **A differential statistic.** At the heights accessible to us, sums over a few thousand zeros fluctuate by several per cent around their expected values. For $\zeta(s)$, the ratio of $\sum 1/|\zeta'(\rho)|^2$ over consecutive windows of 3000 zeros to $\frac{3}{\pi^3}\Delta T$ ranges from 0.907 to 1.088 in the first 27000 zeros, with no trend in the height; most windows fall below 1. This is the behaviour expected from heavy-tailed summands (Section 6). Rather than compare absolute averages with the conjectured constants, we compare each L-function with $\zeta(s)$ at the same local density $D = \log(q\gamma/2\pi)$. We split the zeros of $L(s,\chi)$ into ten bins at the deciles of D and define

$$R = \frac{\operatorname{mean}_\chi(v)}{\sum_b \frac{N_b}{N}\,\operatorname{mean}_{\zeta,\,b}(v)},$$

where N_b is the number of zeros of $L(s,\chi)$ in bin b and $\operatorname{mean}_{\zeta,b}(v)$ is the average of v over the zeros of ζ whose density $\log(\gamma/2\pi)$ lies in the same bin; R_8 is defined in the same way with w. Conjectures (2) and (3) predict

$$R = \prod_{p|q}\frac{p}{p+1}, \qquad R_8 = \frac{\varphi(q)}{q}.$$

We designed this statistic in the expectation that $L(s,\chi)$ and $\zeta(s)$ share finite-height corrections that depend on the height mainly through the density. The heuristic of Section 6 suggests that there are no such corrections of size $T/\log T$; the comparison with ζ then mainly calibrates the typical behaviour of heavy-tailed sums at the same density, and it adds the fluctuations of the ζ-sum to the uncertainty, which the bootstrap below takes into account. The absolute ratios are reported in Section 4 for comparison.

2.3. **Alternatives.** For each test we also record two alternatives: H_0, no arithmetic factor ($R = 1$); and, for (2), the factor $\prod_{p|q} p^2/(p^2-1)$ obtained by simply removing the Euler factors at $p \mid q$ from $1/\zeta(2)$; for (3), the factor of (2), which tests whether the two moments require different factors.

2.4. **Uncertainty.** The terms $1/|L'(\rho,\chi)|^2$ have a heavy tail produced by close pairs of zeros: the largest 1% of the terms carry 12–23% of the sums. We use 95% percentile intervals from a block bootstrap (blocks of 100 consecutive zeros, resampled independently for $L(s,\chi)$ and $\zeta(s)$, 2000 replicates). Because the bootstrap may under-cover with heavy tails, we also recompute R after removing the five largest terms of $L(s,\chi)$.

2.5. **Pre-registration.** For each group of tests, the predictions, alternatives, statistics, bootstrap settings and decision rules were written down and fixed (with SHA-256 hashes) before the derivatives were computed. A test is counted as favourable if the predicted value lies in the interval and H_0 does not.

3. Data

3.1. **Zeros of $\zeta(s)$.** We use the first 27000 zeros ($\gamma \le 23476.2$) from Odlyzko's table `zeros1` (declared accuracy $3 \cdot 10^{-9}$). The tests for $q = 4, 5$ use the first 20000 zeros, and those for $q = 8, 12$ use 27000, so that the densities of the reference cover those of the L-functions.

3.2. **Zeros of $L(s, \chi)$.** We use the real primitive characters $\chi_{-4} = (\frac{-4}{\cdot})$ (odd), $\chi_5 = (\frac{\cdot}{5})$, $\chi_8 = (\frac{8}{\cdot})$ and $\chi_{12} = (\frac{12}{\cdot})$ (even). Zeros were located by sign changes of the real function $Z_\chi(t) = e^{i\theta_\chi(t)} L(\frac{1}{2} + it, \chi)$, with L evaluated by Hurwitz zeta sums in `mpmath` at 30 digits, a scan step of at most $1/8$ of the mean spacing, and refinement to 10^{-10}. Each list was checked for (V1) agreement of the count with $\theta_\chi(T)/\pi$ at the end points, (V2) agreement with an independent rescan at half the step, (V3) a second evaluation at random zeros, and (V4) reality of Z_χ.

Sign-change scans lose pairs of very close zeros. This happened: the original lists for χ_{-4} and χ_5 failed V1/V2 (deficits of 14 and 6 zeros in 12000), and the list for χ_8 failed V2 (2000 found against 2002 in the rescan). In every case the missing zeros were recovered by rescanning all gaps larger than 1.6 mean spacings with a step of $1/32$ of the mean spacing; they were always close pairs (for χ_8, the pair 1470.2269, 1470.2791). The corrected lists pass all checks, and the corrected χ_8 list has exactly the 2002 zeros found by the rescan. We use the zeros 1–3000 for $q = 4, 5$ ($\gamma \le 2891.3$ and 2808.1) and 1–2000 for $q = 8, 12$ ($\gamma \le 1856.6$ and 1764.1).

3.3. **Derivatives and $L(2\rho, \chi^2)$.** Since $|Z'_\chi(\gamma)| = |L'(\rho, \chi)|$ at a zero, we computed $|\zeta'(\rho)|$ with `mpmath.siegelz` and $|L'(\rho, \chi)|$ by a central difference of Z_χ with step 10^{-6} at 20 digits (relative error below $4 \cdot 10^{-10}$ against step 10^{-5} at 30 digits). For real χ, χ^2 is the principal character, so $L(2\rho, \chi^2) = \zeta(1 + 2i\gamma) \prod_{p|q} (1 - p^{-1-2i\gamma})$.

4. Results

q	moment	R measured	95% interval	predicted	alternatives excluded
4	J	0.655	$[0.592, 0.715]$	2/3	1, 4/3
5	J	0.827	$[0.753, 0.901]$	5/6	1, 25/24
8	J	0.687	$[0.605, 0.784]$	2/3	1, 4/3
12	J	0.501	$[0.445, 0.555]$	1/2	1, 3/2
4	K	0.490	$[0.430, 0.549]$	1/2	1, 2/3
5	K	0.753	$[0.673, 0.835]$	4/5	1 (not 5/6)
8	K	0.558	$[0.450, 0.684]$	1/2	1 (not 2/3)
12	K	0.340	$[0.301, 0.381]$	1/3	1, 1/2

Table 1. Differential statistics R (for J_χ) and R_8 (for K_χ). Removing the five largest terms gives 0.633 ($q = 4$) and 0.794 ($q = 5$) for J, without changing any conclusion.

In all eight tests the predicted factor lies inside the interval and H_0 is excluded (Table 1). In six of them the other alternative is excluded as well. In particular, for $q = 4$ and $q = 12$ the two moments require their own factors: exchanging the factors of (2) and (3) is excluded. The

characters χ_8 and χ_{-4} have the same predicted factors but opposite parity, and their measured values are compatible.

For comparison, the absolute ratios $J_\chi(T)/(\frac{A_q}{2\pi}T)$ are 0.937, 0.937, 0.970 and 0.966 for $q = 4, 5, 8, 12$, and $K_\chi(T)/(\frac{\varphi(q)}{q}\frac{T}{2\pi})$ are 0.921, 0.871, 1.023 and 0.949. The corresponding ratios for $\zeta(s)$ over the same numbers of zeros are 0.920 and 0.932 (3000 zeros) and 0.949 and 0.959 (2000 zeros). All these values lie within a few per cent of 1, whereas the alternative without arithmetic factor would give absolute ratios between 1.2 and 3.

5. A heuristic check of the arithmetic factors

The constants in (2) and (3) can also be recovered from the random matrix model of Hughes, Keating and O'Connell [6], independently of the mollifier heuristic. For $\zeta(s)$ that model predicts $\sum_{0<\gamma\le T}|\zeta'(\rho)|^{2k} \sim \frac{T}{2\pi}\frac{G^2(k+2)}{G(2k+3)}a(k)(\log T)^{k(k+2)+1}$, where G is Barnes' function and $a(k) = \prod_p(1-\frac{1}{p})^{k^2}\,\mathbb{E}_\theta|1-p^{-1/2}e^{i\theta}|^{-2k}$; for $k=-1$ this gives (1). For a primitive character χ modulo q we keep the same random matrix factor and replace the local average by $\mathbb{E}_\theta|1-\chi(p)p^{-1/2}e^{i\theta}|^{-2k}$, which equals 1 when $p \mid q$, while the factor $(1-\frac{1}{p})^{k^2}$ is kept for every prime. For $k=1$ this reproduces the factor $\varphi(q)/q$ in the mean square $\int_0^T |L(\frac12+it,\chi)|^2\,dt$, as follows from the approximate functional equation and the mean value theorem. For $k=-1$ it gives

$$\prod_{p\nmid q}\Big(1-\frac{1}{p^2}\Big)\prod_{p\mid q}\Big(1-\frac{1}{p}\Big) = \frac{6}{\pi^2}\prod_{p\mid q}\frac{p}{p+1} = A_q.$$

In the same way, the local average of $|L_p(2\rho,\chi^2)|^2|L_p(\rho,\chi)|^{-2}$ equals $1/(1-\frac1p)$ for $p\nmid q$ and 1 for $p\mid q$, which gives the factor $\varphi(q)/q$ of (3). Thus two independent heuristics lead to the same constants. The alternatives excluded by the data also have a simple meaning in this picture: the factor $\prod_{p\mid q}p^2/(p^2-1)$ corresponds to dropping $(1-\frac1p)$ at $p\mid q$, and exchanging the factors of (2) and (3) corresponds to using the local average of one moment for the other. This is a heuristic consistency check and says nothing about lower-order terms; we do not claim it is new.

6. The ratios conjecture and the error term

The recipe of the ratios conjecture [1, 2] also gives (2) and (3), together with an error term. The representation of discrete moments used in [2, Section 7], with the weight ζ'/ζ, is not suitable here: for negative moments the auxiliary poles cancel the contribution of the zeros. Instead, for $\operatorname{Re}\alpha > 0$ small, write

$$J(\alpha) = \sum_{0<\gamma\le T}\frac{1}{\zeta'(\rho)\,\zeta(1-\rho+\alpha)} = \frac{1}{2\pi i}\oint_C \frac{ds}{\zeta(s)\,\zeta(1-s+\alpha)},$$

where C is the rectangle with vertical sides $\operatorname{Re}s = \frac12-\varepsilon$ and $\operatorname{Re}s = \frac12+\varepsilon'$, $0<\varepsilon'<\operatorname{Re}\alpha$, so that only the poles $s=\rho$ lie inside. Under the Riemann Hypothesis, $\sum_{0<\gamma\le T}|\zeta'(\rho)|^{-2} = \lim_{\alpha\to0}\alpha J(\alpha)$. On the right side the integrand is $1/(\zeta(\frac12+\varepsilon'+it)\zeta(\frac12+\alpha-\varepsilon'-it))$, with both shifts having positive real part; since there is no ζ in the numerator, the recipe keeps only the diagonal $\sum_h \mu(h)^2h^{-1-\alpha} = \zeta(1+\alpha)/\zeta(2+2\alpha)$ (this is [2, (2.10)] with the numerator shifts sent to infinity). On the left side the functional equation leaves an unbalanced factor $\chi(s)$, and the recipe discards such terms, which contribute $O(T^{1/2+\varepsilon})$. Hence

$$J(\alpha) = \frac{T}{2\pi}\frac{\zeta(1+\alpha)}{\zeta(2+2\alpha)} + O(T^{1/2+\varepsilon}), \qquad \sum_{0<\gamma\le T}\frac{1}{|\zeta'(\rho)|^2} = \frac{3}{\pi^3}\,T + O(T^{1/2+\varepsilon}).$$

For $L(s,\chi)$ the diagonal becomes

$$\sum_h \frac{\mu(h)^2|\chi(h)|^2}{h^{1+\alpha}} = \frac{L(1+\alpha,\chi_0)}{L(2+2\alpha,\chi_0)},$$

with χ_0 the principal character modulo q, and $\alpha L(1+\alpha,\chi_0)/L(2+2\alpha,\chi_0) \to A_q$. For K_χ we use $L(2s,\chi^2)/L(s,\chi) = \sum_h \lambda(h)\chi(h)h^{-s}$; the diagonal is $L(1+\alpha,\chi_0)$, and $\alpha L(1+\alpha,\chi_0) \to \varphi(q)/q$. Thus the recipe gives

$$J_\chi(T) = \frac{A_q}{2\pi}\,T + O(T^{1/2+\varepsilon}), \qquad K_\chi(T) = \frac{\varphi(q)}{q}\,\frac{T}{2\pi} + O(T^{1/2+\varepsilon}),$$

which answers heuristically the question raised in [9]. In particular, no lower-order terms of size $T/\log T$ are predicted. The summands, however, have heavy tails: since the nearest-neighbour spacing density vanishes quadratically, $\mathbb{P}(|\zeta'(\rho)|^{-2} > x)$ decays like $x^{-3/2}$, so the summands have infinite variance, and a sum over a few thousand zeros typically falls somewhat below its expected value. This is consistent with the window ratios quoted in Section 2.2 and with the absolute ratios of Section 4. Like the previous section, this is a short heuristic computation, and we do not claim it is new.

7. Discussion and limitations

The data support the arithmetic factors in (2) and (3) for four real characters, including a composite conductor. The evidence is limited in several ways: the heights are low ($\gamma < 3000$); only real characters of small conductor were used; the reference set of ζ-zeros is shared between tests; the bootstrap intervals may be optimistic because of heavy tails; and the differential statistic compares two heavy-tailed sums, whose typical behaviour at the same density of zeros we assume to be comparable.

The recipe of Section 6 predicts that the only term of order T is the main term. Testing this against a possible term of size $T/\log T$ requires many more zeros, because of the heavy tails; a natural test would use fresh zeros at larger heights together with a random-matrix model of the fluctuations of the sums. Complex characters and larger conductors are natural further tests of the arithmetic factors.

8. Literature search

We searched zbMATH Open, the arXiv, and the works citing Gonek's paper on negative moments (via OpenAlex) for numerical computations of $L'(\rho,\chi)$ at zeros of Dirichlet L-functions or of the sums J_χ, K_χ, and found none; the only numerical study of the corresponding moments of ζ that we found is [5]. MathSciNet was not available to us.

Data and code

The zeros, derivatives, pre-registration documents and analysis code, including a script that reproduces Table 1, are available at `https://doi.org/10.5281/zenodo.23020957`. All computations were run on a single laptop with two cores; the computation and verification of the zeros of the L-functions dominated the cost.

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